Yinuo (Nancy) Xu

PhD Candidate, University of Michigan · LLM Personalization & Pluralistic Alignment

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Education

University of Michigan, Ann Arbor — *2023 – Expected 05/2028*
Ph.D. in Information (GPA: 3.80/4.00) — Advisor: David Jurgens

University of California, Berkeley — *B.A. Data Science & Statistics (GPA: 3.83/4.00), 2023*

Publications (Selected)

- **Xu, Y.**, Adams, J., Hu, B., Kumar, S., Jurgens, D., & Basharat, A. (2026). *VISTA: Steering LLMs with latent user values inferred from a few survey answers.* Under review (ARR).
- **Xu, Y.**, & Jurgens, D. (2026). *Beyond consensus: Perspectivist modeling and evaluation of annotator disagreement in NLP.* EMNLP 2026 (Oral).
- **Xu, Y.**, Derricks, V., Earl, A., & Jurgens, D. (2026). *Modeling annotator disagreement with demographic-aware experts and synthetic perspectives.* ACL 2026 (Oral).
- **Xu, Y.**, Chen, H., Rakshit, S., *et al.*, & Jurgens, D. (2025). *Causally modeling the linguistic and social factors that predict email response.* NAACL 2025. (Equal contribution)

Relevant Experience

Research & Development Intern, Kitware — *May–Aug 2026*

Project: VISTA: Steering LLMs with Latent User Values Inferred from a Few Survey Answers [arXiv]

- Led development of VISTA, which infers a probabilistic user-value profile from survey answers (variational encoder) and maps it via a hypernetwork to steering vectors and soft prompts to steer a frozen Llama-3-8B for personalized alignment in choice selection and free-text generation (soft prompts refined with personalized DPO). Matched or beat in-context profile prompting on 3 datasets (up to ∼10 accuracy points) with no added prompt tokens; in free-text generation, using *only* survey answers, outperformed prompting with answers *plus* demographics, with the best net gain across minority- and majority-view users.
- Designed adaptive questioning that personalizes from just 6% of a survey, choosing the most informative questions and stopping automatically while keeping 94% of the full-survey accuracy gain (vs. at most 42% for standard adaptive testing).

Lead Researcher, University of Michigan School of Information — *Aug 2023–Present*

Survey: Perspectivist Modeling of Annotator Disagreement [EMNLP 2026]

- Synthesized 110+ works on annotator disagreement in subjective, safety-relevant tasks (e.g., toxicity) into a unified taxonomy of disagreement sources and a framework of prediction targets and pooling structures, identifying the modeling gaps that motivate Project 2.

Project 1: Modeling Annotator Disagreement with Demographic-Aware Experts [ACL 2026]

- Designed Demographic-Aware Mixture-of-Experts, an interpretable architecture with demographic-aware routing to capture structured variation in annotation behavior & intersectional perspectives.
- Achieved state-of-the-art subgroup-level performance across five benchmark datasets, improving fairness for underrepresented groups under data imbalance and sparse subgroup data.
- Evaluated LLM annotation reliability through zero-shot, few-shot, and LoRA-finetuned experiments, measuring alignment between persona-prompted LLM ratings and human judgments.
- Developed data-efficient pipelines blending real and LLM-generated synthetic annotations with alignment-weighted loss; implemented multi-GPU training workflows for model scaling and reproducibility.

Project 2: Bayesian Hierarchical Adapter Personas

- Designing a hypernetwork-based framework that generates parameter-efficient LoRA adapters from task, data, and annotator-feature interactions, with a three-level Bayesian hierarchy (population, subpopulation, individual) for pluralistic alignment of model behavior with group and individual norms.
- Leading evaluation on a 140K-instance benchmark spanning normative, pragmatic, and preference-judgment tasks, against Jury Learning, Mixture-of-Personas, and sociodemographic prompting baselines.

Project 3: Measuring Racial Disparities in Doctor-Patient Conversations

- Built a zero-shot LLaMA-70B labeling pipeline for doctor communication quality, validated against human ratings; designed a causal mediation model (doctor language → patient behavior → outcome); in prep for *PNAS*.

NLP Intern, Evisort (acquired by Workday) — *May–Aug 2021*

- Trained LightGBM and bi-LSTM NER models (TensorFlow) for contract field extraction, deployed to production; built sensitive-data redaction pipelines with the Google DLP API.

Awards & Leadership Co-Organizer of EMNLP 2027 *Personalization Without Prejudice* workshop; Organizer of NLP@Michigan Day 2026; Organizer of the Michigan NLP Reading Group; Science Communication Fellow (selected as 1 of 20 fellows university-wide); University of Michigan School of Information Merit Fellowship (2023)

Technical Skills Python · PyTorch · Hugging Face Transformers · PEFT / LoRA · vLLM · SLURM · Git · Hypernetworks · Activation Steering · DPO · Mixture-of-Experts · Multi-GPU Training (DDP) · Bayesian & Variational Inference · Causal Inference · LLM Evaluation & Alignment · Optuna